

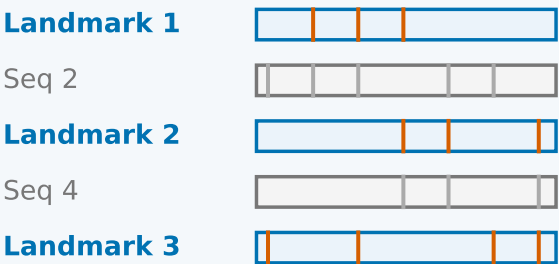
RhizAeon Computational Architecture and Recombination Inference Pipeline

From Multiple Sequence Alignment to Exact Characterization of Mosaic Reticulations in Linear Time

STAGE 1: Metric Embedding

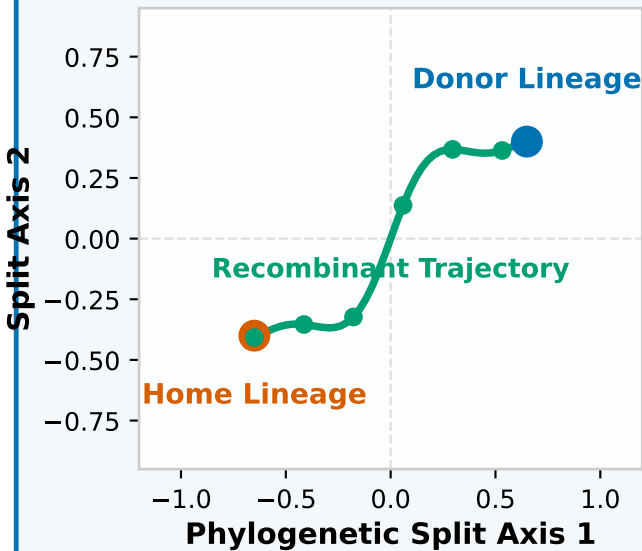
A. Landmark Sampling

- Minimax sampling selects diverse reference genomes.



B. Metric Trajectories

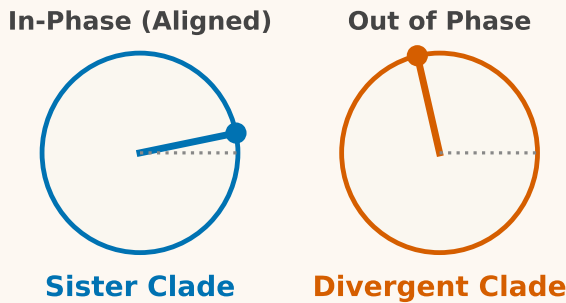
- MDS & Gower projection track continuous spatial trajectories.



STAGE 2: Directed Attention

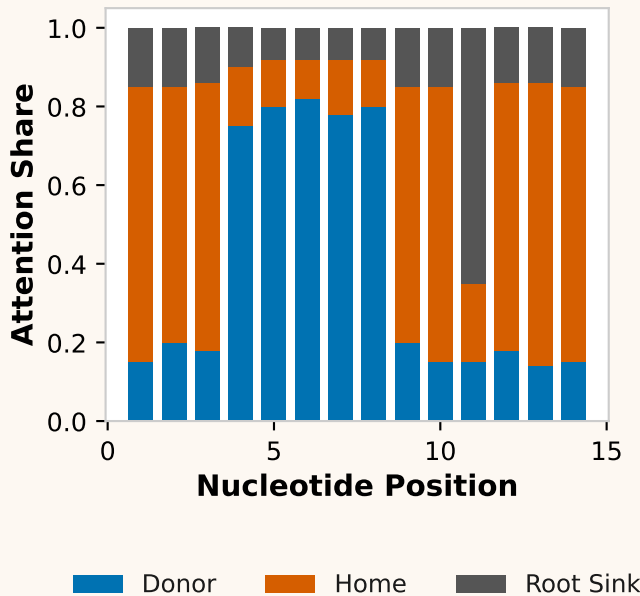
A. Tree-RoPE & Prior

- Tree-RoPE encodes distances; Markov prior elevates key SNPs.



B. Ancestral Root Sink

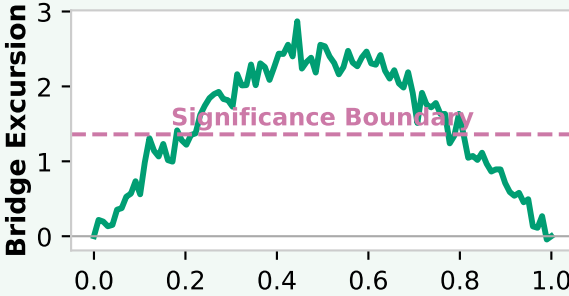
- Origin token absorbs bursts, strictly eliminating false donors.



STAGE 3: Three-Tier Inference

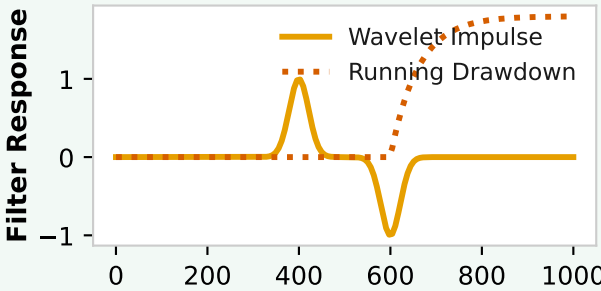
Tier 1: Single Crossovers

- Centered Brownian bridge with analytical boundary test.



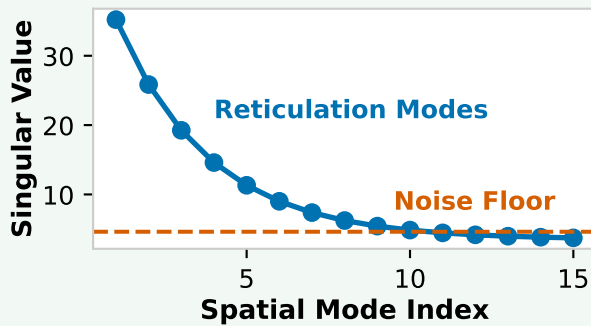
Tier 2: Short Conversions

- Multi-scale wavelets and drawdown isolate short tracts.



Tier 3: Multi-Parent Mosaics

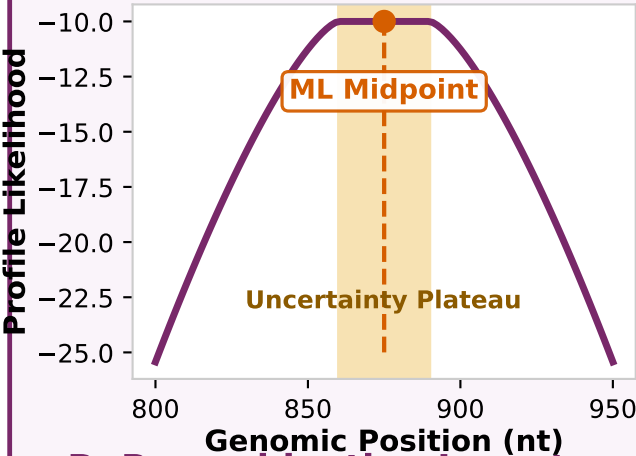
- Force field + SVD mode peeling stops at random noise edge.



STAGE 4: Event Resolution

A. Informational Resolution

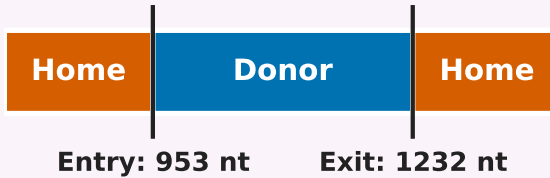
- Profile likelihood measures uncertainty at SNP bounds.



B. Recombination Inventory

- Continuous polarity separates crossovers from cassettes.

Resolved Mosaic Genome Track:



Cassette Event: [953 - 1232 nt]

Donor: Clade B | Home: Clade C